

Unit 6. Special Distributions

Bernoulli, Binomial, Poisson, Uniform, Exponential, Normal, χ^2 , t , F
and the multivariate normal

Introductory Statistics for Econometrics

Universidad Carlos III de Madrid

Adapted from MIT OCW 14.30 (Spring 2009), K. Menzel, Lecture Notes 14, CC BY-NC-SA 4.0.

Roadmap

- 1 Standardisation
- 2 Discrete benchmarks
- 3 Continuous benchmarks
- 4 The normal distribution
- 5 χ^2 , t and F
- 6 Multivariate normal
- 7 Wrap-up

Standardisation

For any random variable X with $\mu = \mathbb{E}[X]$ and $\sigma^2 = \text{Var}(X) > 0$:

$$Z = \frac{X - \mu}{\sigma}, \quad \mathbb{E}[Z] = 0, \quad \text{Var}(Z) = 1.$$

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Why we standardise

- Removes location and scale — shape only.
- Allows universal tables (standard normal Φ).
- Foundation of every z -statistic and t -statistic.

Bernoulli, Binomial, Poisson

- **Bernoulli**(p): $X \in \{0, 1\}$, $\mathbb{P}(X = 1) = p$. $\mathbb{E} = p$, $\text{Var} = p(1 - p)$.
- **Binomial** $B(n, p)$: count of successes in n i.i.d. Bernoulli(p) trials.
 $\mathbb{E} = np$, $\text{Var} = np(1 - p)$.
- **Poisson**(λ): count of rare events, $f(x) = e^{-\lambda} \lambda^x / x!$. $\mathbb{E} = \text{Var} = \lambda$.

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Closure under addition

- Sum of independent Binomials with the **same** p : $B(m, p) + B(n, p) = B(m + n, p)$.
- Sum of independent Poissons: $\text{Poisson}(\lambda_1) + \text{Poisson}(\lambda_2) = \text{Poisson}(\lambda_1 + \lambda_2)$.

Binomial $\rightarrow$ Poisson

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Use this when: large n , small p , modest $\lambda = np$.

- Defaults in a credit portfolio.
- Failures of a manufacturing process.
- Customer arrivals at a call centre.

Uniform and exponential

Uniform $U[a, b]$

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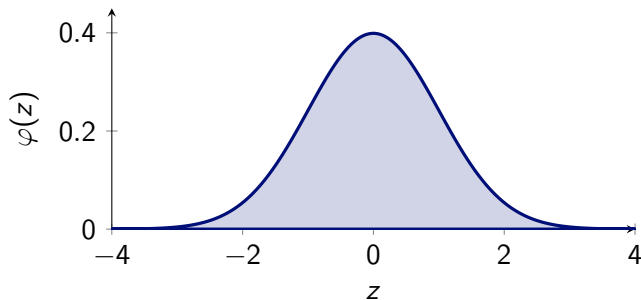
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Memoryless property: $\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t)$.

The exponential is the only continuous distribution with this property. It is the inter-arrival distribution of a Poisson process.

Standard normal $\mathcal{N}(0, 1)$

$$\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad \Phi(z) = \int_{-\infty}^z \varphi(s) ds.$$



Key facts: symmetric around 0; $\Phi(-z) = 1 - \Phi(z)$; no closed form for Φ .

General normal $\mathcal{N}(\mu, \sigma^2)$

Definition. $X = \mu + \sigma Z$ with $Z \sim \mathcal{N}(0, 1)$, so

$$f_X(x) = \frac{1}{\sqrt{2\pi} \sigma} \exp\left[-\frac{(x - \mu)^2}{2\sigma^2}\right].$$

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Two stability properties

- ① **Affine:** $a + bX \sim \mathcal{N}(a + b\mu, b^2\sigma^2)$.
- ② **Independent sums:** $\sum_i X_i \sim \mathcal{N}(\sum_i \mu_i, \sum_i \sigma_i^2)$.

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Result: the normal family is closed under linear maps and under independent addition — a unique combination among standard distributions.

The 68–95–99.7 rule

For $X \sim \mathcal{N}(\mu, \sigma^2)$:

$$\mathbb{P}(|X - \mu| \leq \sigma) \approx 68\%,$$

$$\mathbb{P}(|X - \mu| \leq 2\sigma) \approx 95\%,$$

$$\mathbb{P}(|X - \mu| \leq 3\sigma) \approx 99.7\%.$$

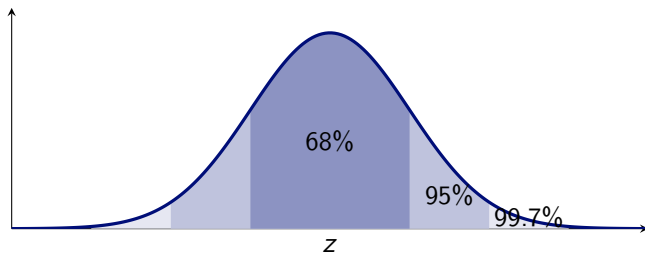
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A handy sanity check whenever you read a confidence interval.

DeMoivre–Laplace: a preview of the CLT

Theorem

$X_n \sim B(n, p)$. Then for every $a < b$,

$$\mathbb{P}\left(a \leq \frac{X_n - np}{\sqrt{np(1-p)}} \leq b\right) \longrightarrow \Phi(b) - \Phi(a).$$

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Why we care.

- Historical ancestor of the central limit theorem (Unit 8).
- Numerically very accurate even for moderate n .
- Lets us replace tedious binomial sums by table lookups in Φ .

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Example. $X \sim B(100, 0.2)$. $\mathbb{P}(X \leq 25) \approx \Phi\left(\frac{25-20}{4}\right) = \Phi(1.25) \approx 0.894$ (exact: 0.913).

Computing normal probabilities

Recipe for $X \sim \mathcal{N}(\mu, \sigma^2)$:

- 1 Standardise: $Z = (X - \mu)/\sigma$.
- 2 Rewrite: $\mathbb{P}(a \leq X \leq b) = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)$.
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Example. $X \sim \mathcal{N}(100, 225)$, find $\mathbb{P}(85 \leq X \leq 120)$:

$$\begin{aligned}\sigma &= 15, \\ \mathbb{P}(85 \leq X \leq 120) &= \Phi(20/15) - \Phi(-15/15) \\ &= \Phi(1.33) - \Phi(-1) \\ &\approx 0.908 - 0.159 = 0.749.\end{aligned}$$

χ^2 , t , F — distributions of statistics

Chi-squared

$Z_1, \dots, Z_k \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$, then $Y = \sum_{i=1}^k Z_i^2 \sim \chi_k^2$. $\mathbb{E}[Y] = k$, $\text{Var}[Y] = 2k$.

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Student's t

$Z \sim \mathcal{N}(0, 1) \perp Y \sim \chi_k^2$, then $T = Z / \sqrt{Y/k} \sim t_k$. For large k , $t_k \rightarrow \mathcal{N}(0, 1)$.

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F-distribution

$Y_1 \sim \chi_{k_1}^2 \perp Y_2 \sim \chi_{k_2}^2$, then $F = (Y_1/k_1)/(Y_2/k_2) \sim F(k_1, k_2)$.

Where they will appear

- χ^2_{n-1} : sampling distribution of $(n-1)S^2/\sigma^2$ for i.i.d. normal samples.
- t_{n-1} : sampling distribution of $(\bar{X} - \mu)/(S/\sqrt{n})$ — foundation of the t -test (Unit 10).
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Lesson

These three families are not interesting in themselves; they matter because they describe the **sampling distribution** of estimators and test statistics that we will compute in Units 9–10.

Multivariate normal $\mathcal{N}_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$

Definition

$\mathbf{X} \in \mathbb{R}^n$ is multivariate normal with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma} \succ 0$ if

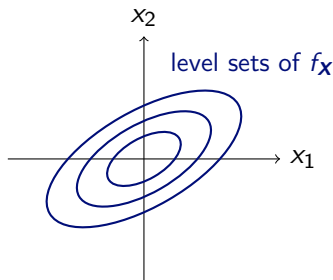
$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{n/2} |\boldsymbol{\Sigma}|^{1/2}} \exp \left[-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right].$$

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Level sets are **ellipses** centred at $\boldsymbol{\mu}$, with axes determined by the eigenvectors of $\boldsymbol{\Sigma}$.

Five essential properties

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- 3 Conditionals are normal. For partition $\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2)$:
 $\mathbf{X}_1 \mid \mathbf{X}_2 \sim \mathcal{N}(\boldsymbol{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{X}_2 - \boldsymbol{\mu}_2), \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})$.

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- 4 Within MVN: $\text{Cov}(X_i, X_j) = 0 \Leftrightarrow X_i \perp X_j$. **AGAIN ONLY UNDER NORMALITY**

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- 5 Standardisation. $\Sigma^{-1/2}(\mathbf{X} - \boldsymbol{\mu}) \sim \mathcal{N}_n(\mathbf{0}, I_n)$, and $(\mathbf{X} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{X} - \boldsymbol{\mu}) \sim \chi_n^2$.

Bivariate normal: the CEF is exactly linear

For $(X, Y) \sim \mathcal{N}_2$ with means μ_X, μ_Y , variances σ_X^2, σ_Y^2 , correlation ρ :

$$\mathbb{E}[Y \mid X = x] = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X),$$

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Bridge to econometrics

The slope is exactly the population OLS coefficient $\beta = \text{Cov}(X, Y)/\text{Var}(X)$ derived in Unit 6 — now **justified** by the assumption of joint normality.

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For non-normal joint distributions, the OLS line is still the best *linear* predictor, but it is no longer the conditional expectation function. This distinction is the start of nonlinear regression and non-parametric methods.

Why MVN is everywhere in econometrics

- **Multivariate CLT**: sums of i.i.d. random vectors converge to MVN. (Unit 8.)
- **OLS estimator**: $\hat{\beta}$ is asymptotically MVN with covariance $\sigma^2(X^\top X)^{-1}$.
- **Joint hypothesis tests** on regression coefficients (the F -test, Wald test) reduce to chi-squared distances under MVN sampling.
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Punchline. Almost everything you will compute in your econometrics course is, asymptotically, multivariate normal.

Take-aways

- 1 Standardisation removes location and scale.
- 2 Bernoulli, Binomial, Poisson on the discrete side; Uniform, Exponential on the continuous side; the exponential is the only memoryless one.
- 3 Normal: closed under affine transforms and independent sums; 68/95/99.7 rule; DeMoivre–Laplace as a preview of the CLT.
- 4 χ^2 , t , F are the sampling distributions of statistics built from i.i.d. normals.
- 5 Multivariate normal: affine, marginals, and conditionals all stay normal; zero correlation $\Leftrightarrow$ independence; the CEF is exactly linear.

Next unit

LLN and CLT — why averages of i.i.d. samples converge to constants and become approximately normal.

Thank you

Questions?

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